

Requirements:

- PySpark: 3.5.3
- Apache Kafka: 4.0.0 (KRaft mode)
- Elasticsearch: 7.17.0 (running on localhost: 9200)
- Kibana: 7.17.0 (running on localhost: 5601)
- Logstash: 7.17.0

Packages:

Spacy

Kafka-python

Steps to run

Terminal Setup Instructions

1. **Terminal 1: Kafka**
Start the Kafka server to enable streaming of data.
2. **Terminal 2: Elasticsearch**
Launch the Elasticsearch service to store and index the extracted named entities.
3. **Terminal 3: Logstash**
Start Logstash with the appropriate configuration file to route data from Kafka into Elasticsearch.
4. **Terminal 4: Kibana**
Launch Kibana to visualize and explore the data stored in Elasticsearch.
5. **Terminal 5: main.py (Producer Script)**
Navigate to the directory containing main.py and run the script. This script produces named entities extracted from live news articles and sends them to Kafka.
6. **Terminal 6: main2.py (Spark Streaming Consumer)**
Navigate to the directory containing main2.py and run it using Spark. This script consumes the data from Kafka, processes it using Spark Streaming, and prepares it for storage in Elasticsearch.